

Thaghrah Cheat Sheet

HTTP:

- http → Show all HTTP traffic
- http.response → Show only HTTP responses
- http.request → Show only HTTP requests
- http.response.code == 200 → Filter successful responses
- http.response.code == 404 → Filter 404 Not Found responses
- http.response.code == 302 → Filter redirect responses
- http.request.method == "POST" → Show POST requests
- http.request.method == "GET" → Show GET requests
- http contains "login" → Find login related packets
- http contains "X-Flag" → Search for custom flag headers
- http contains "Set-Cookie" → Find Set-Cookie headers
- http contains "Cookie:" → Find Cookie headers
- http contains "Location:" → Find redirect Location headers

Important Wireshark Actions:

- Right-click packet → **Follow** → **TCP Stream**
- Right-click packet → **Follow** → **HTTP Stream**
- File → Export Objects → HTTP

TCP:

- tcp → Show all TCP traffic
- tcp.flags.syn == 1 && tcp.flags.ack == 0 → Show only initial SYN packets
- tcp.flags.syn == 1 → Show all packets with SYN flag
- tcp.flags.syn == 1 && tcp.flags.ack == 1 → Show SYN-ACK packets
- tcp.port == 8888 → Filter traffic on port 8888
- tcp.port == 9998 → Filter traffic on port 9998
- tcp.stream eq 0 → Filter a specific TCP stream (change number as needed)
- tcp.flags → Display TCP flags
- tcp.len > 0 → Show packets containing data

Important Wireshark Actions:

- Right-click packet → **Follow** → **TCP Stream**
- Right-click packet → **Follow** → **TCP Stream** → Save as Raw
- Statistics → Conversations → TCP

DNS:

- dns → Show all DNS traffic
- dns.flags.response == 0 → Show only DNS queries
- dns.flags.response == 1 → Show only DNS responses
- dns.qry.name contains "FLAG" → Find queries with FLAG in the domain name
- dns.resp.type == 16 → Filter for TXT record responses
- dns.resp.type == 5 → Filter for CNAME record responses
- dns.count.answers == 2 → Find responses with multiple answer records
- dns contains "TXT" → Search for TXT records
- dns contains "CNAME" → Search for CNAME records

Important Wireshark Actions:

- Expand **DNS** section → **Queries** or **Answers**
- Right-click packet → **Follow** → **UDP Stream** (or TCP if applicable)
- Look under **Answers** section for record types and values

FTP:

- tcp.port == 2121 → Filter FTP control channel traffic
- ftp → Show only FTP commands and responses
- ftp.request.command == "USER" → Find username
- ftp.request.command == "PASS" → Find password
- ftp.request.command == "CWD" → Find Change Working Directory commands
- ftp.request.command == "PASV" → Find Passive mode requests
- ftp.response.code == 227 → Find PASV replies (contains IP and port)
- ftp.request.command == "RETR" → Find file download commands
- ftp.request.command == "RNFR" → Find Rename From commands
- ftp.request.command == "RNTO" → Find Rename To commands

Important Wireshark Actions:

- Right-click packet → **Follow** → **TCP Stream**
- Follow the control channel (port 2121) first, then calculate and follow data ports from PASV replies
- Look for command-response pairs in the followed stream

ICMP:

- `icmp` → Show all ICMP traffic
- `icmp.type == 8` → Filter Echo Request packets (Ping)
- `icmp.type == 0` → Filter Echo Reply packets (Ping response)
- `icmp.type == 3` → Filter Destination Unreachable packets
- `icmp.type == 11` → Filter Time Exceeded packets
- `icmp.type` → Display ICMP Type field
- `icmp.code` → Display ICMP Code field
- `icmp.ident` → Display Identifier field (useful for Echo packets)
- `icmp.seq` → Display Sequence number field

Important Wireshark Actions:

- Expand the **ICMP** section in packet details to see Type, Code, Identifier, and Sequence
- Use **Statistics** → **Packet Lengths** or **Statistics** → **Resolved Addresses** for overview
- Sort packets by time to analyze sequences

SMTP:

- `tcp.port == 2525` → Filter SMTP traffic (common in these challenges)
- `smtp` → Show only SMTP commands and responses
- `smtp.req.command` → Filter SMTP request commands
- `smtp.resp.code` → Filter SMTP response codes
- `smtp contains "EHLO"` → Find EHLO commands
- `smtp contains "HELO"` → Find HELO commands
- `smtp contains "MAIL FROM"` → Find sender information
- `smtp contains "RCPT TO"` → Find recipient information
- `smtp contains "DATA"` → Find start of mail data
- `smtp.resp.code == 354` → Find "Start mail input" reply

Important Wireshark Actions:

- Right-click packet → **Follow** → **TCP Stream** (Essential for all SMTP challenges)
- Look for command-response pairs in the followed stream
- Examine headers and body after the DATA command

Forensics:

- dns → Show all DNS traffic
- http → Show all HTTP traffic
- icmp → Show all ICMP traffic
- tcp.port == 53 → Alternative DNS filter
- http.request.method == "POST" → Find login brute force attempts
- http.response.code == 200 → Successful HTTP responses
- http.response.code == 401 → Failed login attempts
- dns.qry.name → Examine queried domain names
- icmp.type == 8 → ICMP Echo Requests
- frame.time → Sort packets chronologically

Important Wireshark Actions:

- Right-click packet → **Follow** → **TCP Stream / UDP Stream**
- Right-click → **Follow** → **HTTP Stream**
- Examine packet **payload** in hexadecimal pane (for ICMP exfiltration)
- Sort packets by **Time** column for chronological order
- Use **Statistics** → **Conversations** to see multi-protocol activity